2C303 31
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Module - 1

2.)

a.) i) Virtual machine concept is an important part of modern operating systems. Instead of creating multiple machines for performing various tasks for multiple users, we can have a single machine to perform all those tasks as a single machine.

ii.) It is mostly used in very huge servers where their are a lot of clients that need to be provided with services.

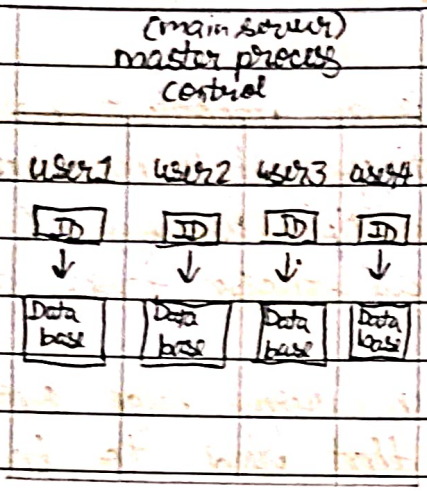
iii.) The virtual machine operating system allows us to create as many seperate "virtual machines" as we need to satisfy the need for every client.

iv.) Each client can get their own personal operating system as if it were their own seperate machine.

v.) This concept of virtual machine operating system is used in many large companies & organisations.

vi.) This system requires a master process control that can create seperate, child processes for every client.

vii.) This system also makes sure that the work of any client system does not affects the other clients.



Each user or client gets a unique ID & the main system needs to handle all the client Data bases carefully.

b.) System calls are basically calls made to the software or operating systems in order to intervene & carry out a specific request or task. After calling the operating system, the computer goes into the Kernel mode. After the request is complete the compater goes back into user mode.

There are many types of system call :-

i.) Process call
ii.) Deletion call
iii.) File management
iv.) Information
v.) Protection
vi.) Communication.

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i) Process Call :- When we want to create a new process or delete an existing process. The user creates a new process can assign attributes to it. or even extract attributes from it. These are Process calls.

ii) File management :- We we want to creat a new file or delete an existing file. The user first needs to creat a new file using file command, can assign attributes to it, can extract or get attributes from it. It plays a major part in file handling.

iii) Deletion :- When we want delete an existing element like process or files etc.

iv) Information :- Allows us to handle information in the device.

v) Protection :- Allows us to create safeguards for important elements, in the device.

vi) Communication :- This helps us to establish communication between different elements in a process or a program, Ex:- Inter process Communication.

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Module - 2

3).

a.) Interprocess communication can be described as communication between 2 processes or the flow or exchange of information between 2 processes. There are many different types of processes, some need to exchange information & some processes do not need to exchange information between each other.

There are 2 ways of exchanging information b/w 2 processes :-

i) Direct communication.
ii) indirect communication.

i) Direct communication :- In direct communication we pass or send the message explicitly between two processes. We can achive this by using the send & receive keywords. Ex :- send(a, message)
receive(b, message).

ii) Indirect communication :- In indirect communication we need to pass the message to the receiving process, by the use of the mailbox. Every process will have a unique mailbox ID which can help it to receive messages. We also use the send & receive keyword here.

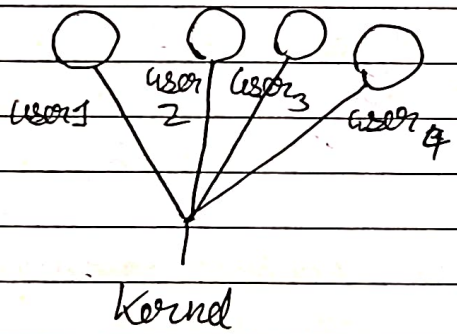
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Message passing in indirect communication works by the use of mail boxes, Each receiver process has a unique mailbox ID which it can use in order to receive messages from the sender processes. We can use the send & receive key words accordingly in order to send or receive messages. Naming, Synch, Buffering

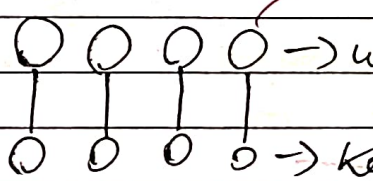
Shared memory :- Many times, Resources are allocated to different processes, but it is not economical as it can clog up the computer memory or main memory & heavily reduce the processing power of the computer. Instead what we can do is share resources like memory between multiple processes. * PTO

b.) i.) many to one :- i.) It has many users node & one @ kernel node

ii.) A blocking function from any user can disrupt other users.



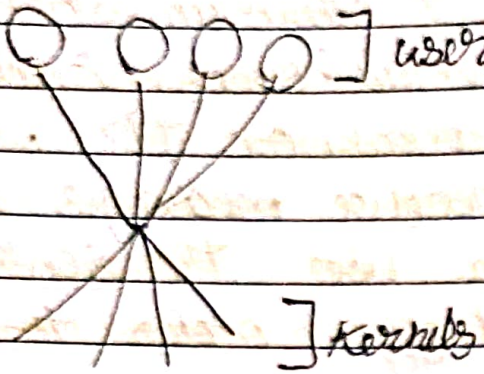
ii.) many to many :- i.) It has every user node connected to a Kernel node



ii.) A blocking function will not disrupt.

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iii.)

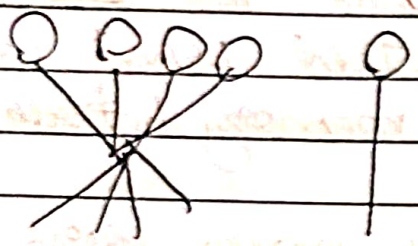


i.) multiple user nodes

ii.) but there are also multiple parallel kernel node

iii.) A blocking function disrupt.

iv.) Mix model :-



i.) A mixed model can be combined with many to many models

Quiz :-

1.) b.) ✗

2.) a.) ✓

3.) b.) ✓

4.) b.) ✓

5.) c.) ✗

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3 a.) PTO

* This way we can also ensure the programing instructions are brief & easy to understand.
Another great advantage of this is that we can Utilise many multiple archives. All of these these things can help us make our instruction more responsive

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